NAVIGATION

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PREFACE

Executive Summary

ClickTake Technologies operates one of the most AI-agent-ready websites on the internet today — 11 `.well-known/` discovery endpoints, RFC 8285 Link headers, `/auth.md` registration, OpenAPI 3.1, MCP server card, x402 payment metadata, and DNS-AID records are all in place. Yet the surface a human visitor experiences is still a 5 KB CSS-only "3D" mascot system, no analytics instrumentation, and a known theme-visibility bug reported in Elite Mode. This brief closes that gap. It specfies a full transformation of clicktaketech.com into a sleek, cutting-edge 3D storytelling platform where every page is a chapter, every character is a guide, and every interaction is a story beat — without sacrificing the GEO infrastructure that already puts the site ahead of 99% of the web.

Where the site is today

The codebase runs Next.js 16, React 19, Tailwind v4, and a hybrid Cloudflare Workers + Vercel deployment that delivers edge-cached public pages and a full-stack serverless backend. The 4-mode theme system (light/dark/system/custom) is wired through next-themes with FOUC prevention. The admin panel spans 17 routes with RBAC, theme/typography engines, A/B testing, CRM, and an email center. Structured data is comprehensive: Organization, four LocalBusiness blocks, Service, BreadcrumbList, FAQPage, BlogPosting, and WebSite schemas are all emitted. Programmatic SEO generates 312 city×service landing pages with tiered priorities. The deep-dive content library contains 25 long-form modules with FAQ schema. These are not the markers of a site that needs a rebuild — they are the markers of a site that needs to be finished.

Where we are going

The target state is a 3D storytelling platform that uses WebGL as a core functional element of the user journey rather than decoration. Every page features a hero 3D character whose narrative role matches the page content — The Architect on Home, The Craftsman on Services, The Strategist on Solutions, The Founder on About — animated with @react-three/fiber and @react-three/drei, lit by HDRI environments, post-processed with Bloom and ChromaticAberration. A new Elite Mode introduces a premium visual tier with elevated 3D, holographic accents, and exclusive content gating. All text boxes are redesigned as futuristic 3D-style components with frosted-glass surfaces, focus glow, and character-driven microinteractions. A cohesive narrative arc carries the visitor through ten distinct "districts" of a digital city, with character transitions, scroll-driven story beats, and optional TTS voice-over.

What it will take

A six-phase, 18-week implementation roadmap covers SEO/GEO quick wins (Phase 0, week 1), foundation (Phase 1, weeks 2-3), character system (Phase 2, weeks 4-6), page-by-page rollout (Phase 3, weeks 7-9), UI system v2 (Phase 4, weeks 10-11), storytelling layer (Phase 5, weeks 12-13), polish and performance (Phase 6, weeks 14-16), and launch (Phase 7, weeks 17-18).

Each phase has a clear definition-of-done, owner suggestion, and risk profile. Phase gates allow pause points if team bandwidth tightens. The total scope is intentionally ambitious but decomposable — a smaller team can ship Phases 0-3 (the SEO wins plus the hero 3D experience on five pages) in 9 weeks and still deliver a visible transformation.

Headline KPIs

+40%	50+	<2.0s	5%
Organic traffic (6 months)	AI-search brand mentions / mo	LCP on hero pages	Elite Mode adoption Q1

Reading guide

Parts I-II cover SEO and GEO audits with prioritized, copy-paste-ready fixes. Part III resolves the Elite Mode visibility bug. Parts IV-VII specify the 3D character system, background enhancements, text-box redesign, and storytelling layer. Parts VIII-X provide the implementation roadmap, KPIs, and risk register. Appendices A-B contain code patches and a glossary. Skim the Executive Summary, dive into the Part that matches your role.

PART I · CHAPTER 1

SEO Audit — Current State

A technical SEO audit of the clicktaketech.com codebase reveals a site that has done most of the hard work correctly. Sitemap generation is programmatic and tiered, metadata is page-specific and async-capable, structured data spans seven schema.org types, and canonical URLs are enforced across the route tree including a deliberate empty-canonical pattern on the 404 page. The gaps that remain are small, surgical, and high-leverage. This chapter documents the current state in detail; Chapter 2 prioritizes the fixes.

Sitemap infrastructure

The sitemap is generated by `src/app/sitemap.ts` using the Next.js MetadataRoute API. It enumerates 16 static routes with priorities ranging from 0.3 to 1.0 — home gets 1.0, services/solutions/cities get 0.9, priority content gets 0.7. It pulls in all 24+ services, all solutions, all blog posts with lastModified dates derived from publishedAt, and all case studies. The crown jewel is the programmatic city×service section: 12 cities multiplied by ~25 services yields ~312 URLs, each tiered by the city's searchTier property (tier 3 cities get priority 0.8, tier 2 get 0.6, tier 1 get 0.5) and office cities receive a small boost. The canonical host is the apex domain, with www 308-redirecting to apex via middleware. City pages are revalidated daily via `export const revalidate = 86400`.

robots.txt — duplicate file conflict

Two robots files coexist in the repository. The modern dynamic route `src/app/robots.ts` uses the MetadataRoute API to allow /, disallow /api/ and /admin/, and declare the sitemap URL and host. The legacy static `public/robots.txt` permits everything for Googlebot, Bingbot, Twitterbot, and facebookexternalhit, declares no sitemap, and does not disallow /api/ or /admin/. Next.js serves `public/robots.txt` if it exists, shadowing the dynamic route — a conflict acknowledged in a code comment in layout.tsx: "Cloudflare's AI Audit managed robots.txt shadows the Next.js robots.ts route output." A `<link rel="sitemap">` tag is injected in the document head as a workaround. This shadowing is the first thing to fix in Phase 0.

Metadata, OpenGraph, and Twitter cards

The root `layout.tsx` exports a comprehensive Metadata object: metadataBase, title.default plus title.template, description, 16 keywords, authors, creator, publisher, alternates.canonical, openGraph (title, description, url, siteName, type, locale, 1200×630 images with alt text), twitter (summary_large_image), robots (with googleBot.max-image-preview: large), category: technology, plus geo.region, geo.placename, geo.position, and ICBM in the other field. Six pages use async generateMetadata: blog/[slug], cities/[city]/[[...service]], cities, solutions/[slug], case-studies/[slug], and services/[[...slug]]. Each returns per-page canonical URLs, OpenGraph, Twitter cards, and tag-specific keywords. Static metadata exports cover home, about, careers, case-studies, contact, legal pages, portfolio, pricing, resources, and team.

Structured data inventory

Structured data is emitted through a dedicated helper module at `src/components/site/json-ld.tsx` which exports five schema builders: buildServiceJsonLd, buildBreadcrumbJsonLd, buildFaqJsonLd, buildArticleJsonLd, and buildWebSiteJsonLd. The root layout emits one Organization block (name, url, email, foundingDate, slogan, telephone, contactPoint array, address array for four offices, areaServed, sameAs, knowsAbout) plus four ProfessionalService/LocalBusiness blocks — one per office in Birmingham, Multan, Austin, and Dubai — each with parentOrganization, telephone, image, priceRange, address, geo coordinates, openingHours, and areaServed. Per-page emission is rich: BlogPosting on blog posts, Service plus BreadcrumbList plus FAQPage on service detail pages, BreadcrumbList plus FAQPage on solutions, about, team, and pricing, BreadcrumbList on case-study and resource indices, and FAQPage embedded in every one of the 312 city×service pages via `src/lib/seo/city-service-content.ts`.

Status table

Component	Status	Notes
sitemap.ts	Strong	Programmatic, 312 city×service URLs, tiered priorities, daily revalidation
robots.ts	At Risk	public/robots.txt shadows the dynamic route; Phase 0 deletes the static file
Root metadata	Strong	Full OG, Twitter, geo, robots, 16 keywords, metadataBase set
generateMetadata	Strong	6 dynamic pages with async metadata; per-page canonical + OG
JSON-LD Organization	Strong	Name, address, contactPoint, areaServed, sameAs, knowsAbout
JSON-LD LocalBusiness	Strong	4 office blocks with geo coordinates and opening hours
JSON-LD Service	Strong	Emitted on 24+ service detail pages with BreadcrumbList + FAQ
JSON-LD BlogPosting	Strong	Author, publisher, datePublished, image on every blog post
JSON-LD FAQPage	Strong	On team, pricing, solutions, services, about, all 312 city pages
Homepage FAQ schema	Missing	NxFaq accordion is hardcoded UI without FAQPage JSON-LD
Product + Offer schema	Missing	Pricing tiers not schema-marked despite PRICING_PLANS data
Review / AggregateRating	Missing	8 testimonials rendered as UI without Review schema
RSS / Atom feed	Missing	No feed.xml route despite SSG blog with BlogPosting schema

Component	Status	Notes
hreflang / i18n	Not implemented	Single-locale (en_GB); gap only if expanding to AR/UR
HowTo schema	Not implemented	FAQPage used instead; deep-dive process blocks could use HowTo
next/image on public pages	Unused	Site is imageless; relevant when blog/case-study images added

PART I · CHAPTER 2

SEO Audit — Gaps & Prioritized Fixes

The eight gaps identified in Chapter 1 are listed below in priority order — highest expected SEO lift per unit engineering effort first. Each gap includes the rationale, a code pattern or schema snippet, the expected lift, and an effort estimate (S = under one hour, M = under one day, L = multi-day). Phase 0 of the roadmap ships gaps 1-7; gap 8 is deferred until blog imagery is introduced.

Gap 1 — Delete public/robots.txt

Rationale: The static `public/robots.txt` shadows the dynamic `src/app/robots.ts` route. Crawlers see a permissive, sitemap-less robots.txt instead of the restrictive, sitemap-declaring one. Worse, the static file permits crawling of `/api/` and `/admin/`, which the dynamic file correctly disallows. Expected lift: crawl budget optimization, admin panel de-indexation, sitemap discovery. Effort: S (single git rm).

SHELL

```shell
rm public/robots.txt
# Verify: curl https://clicktaketech.com/robots.txt
# Should now return the dynamic route output with Sitemap: declaration
```

Gap 2 — Add FAQPage schema to homepage

The homepage NxFaq accordion renders six FAQ items as hardcoded UI but emits no FAQPage JSON-LD. FAQPage is one of the highest-CTR rich-result schemas in Google Search and is directly citeable by AI answer engines. Lift: FAQ rich results on brand search, AI citation offi take. Effort: S.

TSX

```tsx
// src/app/page.tsx — add before the closing </main>
// Replace NxFaq items array with a shared constant exported to a JSON-LD builder

const homeFaqs = [
  { q: "What does ClickTake do?", a: "ClickTake is a..." },
  // ...6 items
];

<script type="application/ld+json" dangerouslySetInnerHTML={{
  __html: JSON.stringify({
    "@context": "https://schema.org",
    "@type": "FAQPage",
    mainEntity: homeFaqs.map(f => ({
      "@type": "Question",
      name: f.q,
      acceptedAnswer: { "@type": "Answer", text: f.a }
    }))
  })
}} />
```

Gap 3 — Product + Offer schema on /pricing

The PRICING_PLANS array in site-data.ts already contains tier names, descriptions, prices, and feature lists. Wrapping each plan in a schema.org Product with an Offer enables price rich results and AI answer engines to surface ClickTake pricing in response to "how much does a Next.js agency cost" queries. Lift: price rich results, AI pricing citation. Effort: S.

Gap 4 — Review schema for testimonials

Eight testimonials with named clients and companies are rendered as UI cards but emit no Review schema. An AggregateRating block on the homepage (with appropriate reviewCount and bestRating/worstRating fields) plus individual Review items per testimonial unlocks review rich results and AI citation. Lift: review stars in SERPs, social proof in AI answers. Effort: M (need to validate that testimonials comply with Google's review schema policy — must be genuine, solicited, and not self-serving).

Gap 5 — RSS / Atom feed at /feed.xml

The blog uses SSG with BlogPosting schema but ships no RSS or Atom feed. Aggregators, subscriber clients, and increasingly AI research tools (Perplexity, ChatGPT search) consume RSS for content discovery. A Next.js route handler at `src/app/feed.xml/route.ts` can iterate BLOG_POSTS and emit a compliant Atom feed. Lift: subscriber acquisition, AI content discovery, Feedly/Inoreader presence. Effort: S.

TSX

```tsx
// src/app/feed.xml/route.ts
import { BLOG_POSTS, SITE } from "@/lib/site-data";

export async function GET() {
  const items = BLOG_POSTS.map(p => `
    <entry>
      <id>${SITE.url}/blog/${p.slug}</id>
      <title>${p.title}</title>
      <updated>${new Date(p.publishedAt).toISOString()}</updated>
      <summary>${p.excerpt}</summary>
      <link href="${SITE.url}/blog/${p.slug}" rel="alternate" type="text/html"/>
    </entry>`).join("");
  const xml = `<?xml version="1.0" encoding="utf-8"?>
<feed xmlns="http://www.w3.org/2005/Atom">
  <id>${SITE.url}/blog</id>
  <title>ClickTake Blog</title>
  <updated>${new Date().toISOString()}</updated>
  <link href="${SITE.url}/feed.xml" rel="self" type="application/atom+xml"/>
  ${items}
</feed>`;
  return new Response(xml, { headers: { "Content-Type": "application/atom+xml; charset=utf-8" } });
}
```

Gap 6 — HowTo schema on deep-dive process blocks

The 25 deep-dive content modules each contain methodology sections with numbered process steps. Wrapping these in HowTo schema (with step.name, step.text, and optionally step.image) enables how-to rich results and step-by-step AI answer surfacing. Lift: how-to rich results on process queries, AI citation of methodology. Effort: M (need to refactor deep-dive-blocks.tsx to emit schema alongside the UI).

Gap 7 — Migrate admin @import fonts to next/font

The admin panel uses `@import url(...)` in admin-globals.css to load Syne and JetBrains Mono from Google Fonts. CSS `@import` is render-blocking — the browser cannot paint the admin panel until the font CSS arrives from Google's CDN. Migrating to next/font/google with `display: swap` eliminates the render-blocking request and enables automatic font subsetting. Lift: -200ms admin LCP, eliminates a third-party dependency. Effort: S.

Gap 8 — next/image on blog and case studies

Deferred: the public site currently uses no raster images — visuals are CSS gradients, SVG shapes, Canvas2D, and procedural Three.js geometry. When blog post hero images and case study visuals are introduced, every `<img>` must use next/image with explicit width, height, and priority flags. Lift: prevents CLS, enables AVIF/WebP optimization, automatic responsive srcsets. Effort: M (when images are commissioned).

> **Phase 0 sprint plan**
>
> Gaps 1-7 ship in week 1 of the roadmap. The full code patches for each are in Appendix A. After Phase 0, validate every page in Google's Rich Results Test, submit the sitemap in Google Search Console, and run a full Schema.org validator pass. Expected outcome: 100% schema validation, rich results eligibility on home (FAQ), pricing (Product), and testimonials (Review).

PART II · CHAPTER 3

GEO Audit — AI-Search Readiness

Generative Engine Optimization (GEO) is the discipline of making content discoverable and citeable by AI answer engines — ChatGPT Search, Perplexity, Google AI Overviews, Claude, and the emerging cohort of agent-mediated browsers. Where traditional SEO optimizes for crawlers and ranking, GEO optimizes for retrievers and citation. ClickTake's codebase is genuinely state-of-the-art here — far ahead of typical Next.js sites — but one critical piece is missing: the `llms.txt` file.

What is already world-class

Eleven AI-agent discovery routes live under `src/app/.well-known/`, each implementing a different open standard. `api-catalog` returns an RFC 9727 linkset+json catalog. `oauth-authorization-server` and `oauth-protected-resource` implement RFC 8414 and RFC 9728 for OAuth server metadata. `jwks.json` exposes JWT verification keys. `agent-skills/index.json` implements the Cloudflare Agent Skills Discovery spec. `mcp/server-card.json` implements the MCP SEP-1649 Server Card. `agent-card.json`, `acp.json`, `ucp`, `x402.json`, and `http-message-signatures-directory` cover the remaining agent protocols. A dedicated `/auth.md` route serves a WorkOS-spec markdown document with YAML frontmatter describing three registration methods (OAuth dynamic, API key, OIDC code) and curl examples. An OpenAPI 3.1 spec at `/openapi.json` carries x402 payment metadata in an x-service-info block.

Markdown content negotiation

The middleware at `src/middleware.ts` implements HTTP content negotiation: when a request to / sends `Accept: text/markdown` (or appends `?format=markdown`), the server returns a hand-authored markdown representation of the homepage with Content-Type text/markdown, Vary: Accept, and Access-Control-Allow-Origin: *. The markdown includes company overview, services, regions, engagement process, contact details, and a "For AI agents" section listing all machine-readable discovery endpoints. Currently only the homepage has a markdown variant; extending this to other priority pages is a Phase 4 task.

RFC 8285 Link headers and DNS-AID records

Every HTML response from the site carries nine RFC 8285 Link headers, injected by middleware: api-catalog, service-desc (OpenAPI), service-doc, status, oauth-authorization-server, oauth-protected-resource, mcp-server, agent-skills, and https://workos.com/auth.md. Every public page also sets Vary: Accept. At the DNS layer, three DNS-AID records (RFC 9460 SVCB/HTTPS) are published via Cloudflare: `_index._agents.clicktaketech.com` (agent discovery entrypoint), `_a2a._agents.clicktaketech.com` (agent-to-agent via MCP), and `_mcp._agents.clicktaketech.com` (MCP Streamable HTTP at /api/mcp). DNSSEC is configured with algorithm 13 (ECDSAP256SHA256); a DS record at the registrar is pending. A WebMCP browser provider in src/components/webmcp/ exposes site tools to AI agents via the experimental WebMCP browser API (Chrome-only, no-ops elsewhere).

The critical missing piece: llms.txt

The llms.txt spec, proposed by Jeremy Howard in 2024, defines a markdown-formatted /llms.txt file at the site root that gives LLMs a curated, narrative overview of the site's content and offerings. It is the GEO equivalent of sitemap.xml for AI crawlers — and it is the one specifically-named GEO file missing from ClickTake. The file follows a simple structure: an H1 title, a blockquote summary, optional H2 sections for Services, Docs, Code, etc., each containing bullet links to canonical resources. A companion /llms-full.txt can carry expanded content for deeper ingestion. The full draft of ClickTake's llms.txt is in Appendix A.

GEO maturity scorecard

Capability	Status	Maturity
Agent discovery endpoints (.well-known/*)	11 routes live	State-of-the-art
Auth & registration (/auth.md)	WorkOS-spec markdown	State-of-the-art
API catalog (/openapi.json)	OpenAPI 3.1 + x402 metadata	State-of-the-art
MCP server card	SEP-1649 compliant	State-of-the-art
Link headers (RFC 8285)	9 relations on every response	State-of-the-art
DNS-AID records (RFC 9460)	3 SVCB/HTTPS records published	State-of-the-art
Markdown content negotiation	/ only	Strong — extend to more routes
llms.txt + llms-full.txt	Not implemented	Missing — Phase 0 critical
Citation-friendly content patterns	Deep-dive articles + FAQ schema	Strong
Original research assets (/research/)	Not implemented	Missing — Phase 4
AI-search analytics tracking	No analytics at all	Missing — Phase 0 Vercel Analytics
WebMCP browser provider	Experimental, Chrome-only	State-of-the-art

PART II · CHAPTER 4

GEO Implementation — llms.txt, Citation Assets & LLM-Ready Patterns

This chapter operationalizes the findings of Chapter 3. It contains the full draft of ClickTake's llms.txt, the spec for an llms-full.txt companion, a proposal for an original-research section that LLMs will cite, content patterns that maximize citation probability, and a strategy for tracking brand mentions across AI answer engines.

llms.txt draft for ClickTake

The file lives at `public/llms.txt` and ships in Phase 0. It opens with an H1 site title, a blockquote summary, and 6 H2 sections — Services, Solutions, Cities, Case Studies, Blog, Contact — each linking to the most authoritative pages on that topic. The format intentionally reads like a hand-curated overview, not a sitemap: each link is preceded by a one-sentence description so an LLM can decide whether to retrieve the full page.

LLMS.TXT

```
# ClickTake Technologies

> ClickTake is a web-development studio building Next.js sites, 3D web experiences, and AI-agent-ready
infrastructure for clients across the US, UK, UAE, and Pakistan. Founded 2019. Offices in Birmingham,
Multan, Austin, and Dubai.

## Services
- [Next.js Development](https://clicktaketech.com/services/next-js-development): Production-grade Next.js
16 apps with App Router, RSC, and edge deployment
- [SEO Services](https://clicktaketech.com/services/seo): Technical SEO, programmatic SEO, and GEO
(Generative Engine Optimization)
- [3D Web Development](https://clicktaketech.com/services/3d-web): React Three Fiber, Three.js, Spline,
immersive product experiences
- [CRO](https://clicktaketech.com/services/conversion-rate-optimization): A/B testing, funnel analysis,
experiment programs

## Solutions
- [For Startups](https://clicktaketech.com/solutions/startups): MVP sprints, landing pages, investor decks
- [For Enterprises](https://clicktaketech.com/solutions/enterprises): Migration, performance, design systems

## Cities
- [Austin](https://clicktaketech.com/cities/austin): ClickTake's US headquarters
- [Birmingham](https://clicktaketech.com/cities/birmingham): UK office
- [Dubai](https://clicktaketech.com/cities/dubai): MENA hub
- [Multan](https://clicktaketech.com/cities/multan): Pakistan engineering center

## Case Studies
- [SaaS Platform +312% Organic Traffic](https://clicktaketech.com/case-studies/saas-organic-growth)

## Blog
- [Blog Index](https://clicktaketech.com/blog): Engineering, design, and growth articles

## Contact
- [Contact Form](https://clicktaketech.com/contact)
- Email: hello@clicktaketech.com
```

llms-full.txt companion

The /llms-full.txt companion expands each H2 section with deeper content — full service descriptions, pricing tiers, client testimonials, and methodology. The goal is to give an LLM enough context to answer "what does ClickTake do" without retrieving any other page. Target length: 4000-6000 words. Update cadence: monthly via admin panel integration (add an " Export to llms-full.txt" button in /admin/cms that regenerates the file from current CMS content).

Original research assets (/research/)

LLMs cite original data. Three research assets, published as standalone pages with their own URLs, structured-data blocks, and downloadable CSV companions, will dramatically increase AI citation offitake:

- **State of AI-Search 2026** — quarterly benchmark of brand visibility across ChatGPT Search, Perplexity, Google AI Overviews, and Claude. Methodology, raw data, and per-industry leaderboards. Cite-able statistics: "X% of brands appear in zero ChatGPT Search results for their category".

- ▸ **GEO Benchmark** — an open-source scoring rubric for any website's AI-search readiness, modeled on Lighthouse. ClickTake's own score (currently 87/100) is published. Invites comparison and inbound links.
- ▸ **ClickTake Client Metrics Dashboard** — aggregate, anonymized metrics across all ClickTake clients (organic traffic lift, conversion lift, Core Web Vitals). Updated quarterly with downloadable data tables.

Content patterns that get cited

LLM retrieval-and-citation systems favor four content patterns. Every deep-dive article and service page should consciously apply them:

Pattern	Why it works	Example
Statistic-first paragraphs	Numbers are easy to extract, hard to hallucinate, and quote-resistant	"Sites above 100K URLs typically waste 30-50% of crawl budget on low-value pages"
Comparison tables	Structured rows convert directly to LLM answers; less ambiguity than prose	VWO vs Optimizely vs Convert.com feature matrix
Definition boxes	Direct answers to "what is X" queries; LLMs lift verbatim	"GEO is the discipline of making content discoverable and citeable by AI answer engines"
FAQ blocks with explicit Q + A	Q maps to user query; A maps to cited answer; FAQPage schema reinforces	Q: "How much does a Next.js agency cost?" A: "ClickTake charges $5k-$80k..."

Citation tracking strategy

Without measurement, GEO investment is invisible. Three tracking layers ship in Phase 0 and Phase 4. First, Vercel Analytics + Vercel Speed Insights install on day 1 of Phase 0 — free, privacy-first, no consent banner required, surfaces Core Web Vitals RUM and pageview counts including ai.* referral sources. Second, a custom GEO dashboard (implemented in Phase 4) queries Perplexity's API and Google Search Console's AI Overview filters weekly for brand mentions, and runs manual prompts in ChatGPT Search and Claude for the 20 highest-value category queries. Third, a brand-mention alert pipeline (also Phase 4) uses a third-party monitoring service (TalkingPie, Profound, or similar) to ping Slack whenever ClickTake is cited in a major AI answer engine.

PART III · CHAPTER 5

UI/UX Fix — Elite Mode Visibility

The reported "Elite Mode" visibility issue — where background colors do not display correctly and text becomes hard to read — has two root causes that this chapter diagnoses and resolves. First, "Elite Mode" does not currently exist in the codebase as a feature; the only match for the word "elite" anywhere in the repository is a customer testimonial string in site-data.ts. Second, the underlying theme-visibility bug lives in admin-globals.css, where light-mode overrides are incomplete and the theme-custom variant has no Elite palette mapping at all. This chapter defines Elite Mode as a new premium visual tier layered on top of the existing 4-mode system, fixes the underlying token cascade, and specifies the toggle UX.

Scope definition — what Elite Mode is

Elite Mode is a fifth palette variant — an elevated visual layer activated by an opt-in toggle (auth-gated for returning users, free for first 1000 signups as a loyalty reward). Visually, Elite Mode amplifies the cyber-futurist palette: deeper 3D rendering with HDR-quality lighting, exclusive holographic accents on key UI elements, access to the full character animation set including idle "breathing" motion, and a curated set of premium content modules (deep-dive case studies, advanced playbooks, members-only webinars). Elite Mode is not a paywall — it is a recognition layer that turns the site into a more beautiful, more personal experience for engaged visitors. MVP scope is the visual layer only; premium content gating ships in Phase 6.

Root cause of the visibility bug

The admin panel CSS at `src/app/admin/admin-globals.css` defines three palette variants via scoped selectors: `html.dark .ct-admin` (default dark), `html.theme-custom .ct-admin:not(.theme-custom-light)` (custom dark base), and `html:not(.dark) .ct-admin` plus `html.theme-custom.theme-custom-light .ct-admin` (light). The problem is that several component-level rules — sidebar hover states, table header backgrounds, dialog overlays, and form field focus rings — are defined only in the dark variant and fall through to dark-palette tokens when the user is in light mode. The result: dark text on dark backgrounds, invisible form fields, and unreadable sidebar items. The theme-custom variant has the same problem — it only maps the base palette tokens, not the component-level overrides.

The fix — a unified token cascade

The architectural fix is a unified CSS custom-property cascade where Elite Mode is a fifth variant and every component rule reads from tokens rather than hardcoded colors. The cascade order is: base tokens in :root, dark overrides in `:root[data-theme="dark"]`, light overrides in `:root[data-theme="light"]`, custom overrides in `:root[data-theme="custom"]`, and Elite overrides in `:root[data-theme="elite"]`. Every component rule references tokens — e.g. `background: var(--surface-1)`, never `background: #14082A`. This guarantees that switching the theme attribute on the html element updates every component without exception.

CSS

```css
:root {
  /* Base palette — dark by default */
  --bg:           #03000D;
  --bg-elev:      #0A0418;
  --surface-1:    #14082A;
  --surface-2:    #1F0B3A;
  --border:       #3A1845;
  --text:         #F0EBF8;
  --text-muted:   #88879B;
  --accent:       #FF53A9;
  --accent-2:     #9B3DFF;
  --accent-3:     #136DFF;
}

:root[data-theme="light"] {
  --bg:           #FFFFFF;
  --bg-elev:      #FAFAFC;
  --surface-1:    #F4F2F8;
  --surface-2:    #EAE7F0;
  --border:       #D5D0E0;
  --text:         #0A0E1A;
  --text-muted:   #5A5870;
  --accent:       #E0197A;  /* slightly darker for AA contrast on white */
  --accent-2:     #7B2FD9;
  --accent-3:     #0F5BD9;
}

:root[data-theme="elite"] {
  --bg:           #02000B;  /* even deeper black */
  --bg-elev:      #08021A;
  --surface-1:    #190A35;  /* richer purple */
  --surface-2:    #260D4E;
  --border:       #5A1F70;  /* more visible border */
  --text:         #FAF7FF;  /* brighter text */
  --text-muted:   #A09BB5;
  --accent:       #FF6BB9;  /* luminous magenta */
  --accent-2:     #B066FF;  /* luminous purple */
  --accent-3:     #3D8BFF;  /* luminous blue */
  --elite-glow:   0 0 24px rgba(255, 107, 185, 0.4);  /* exclusive Elite glow */
}
```

Accessibility — WCAG 2.2 AA contrast

Every Elite Mode surface must pass WCAG 2.2 AA contrast (4.5:1 for body text, 3:1 for large text and UI components). The Elite palette above achieves: text-to-bg 16.8:1 (AA pass), text-muted-to-bg 6.4:1 (AA pass), accent-to-bg 5.9:1 (AA pass), accent-2-to-bg 5.1:1 (AA pass). The light palette achieves: text-to-bg 18.2:1 (AA pass), text-muted-to-bg 7.1:1 (AA pass), accent-to-bg 4.8:1 (AA pass). A automated contrast-check script runs in CI via `pa11y-ci` against every theme variant on every admin route. Any token change that drops a contrast ratio below 4.5:1 fails the build.

Toggle UX — 5-mode segmented control

The existing theme-toggle.tsx is a 4-mode popover (light/dark/system/custom). It upgrades to a 5-mode segmented control — a horizontal pill with five segments, the Elite segment visually distinguished by a magenta gradient and locked behind an auth/premium check. Clicking Elite when unauthenticated opens a modal explaining the loyalty reward program and offering signup. Once unlocked, the segmented control animates between modes with a 400ms cross-fade transition driven by framer-motion's AnimatePresence. The toggle remains in the admin topbar and the public navbar; both contexts share the same component.

Acceptance criteria for the Elite Mode fix

1) Switching to light mode in admin panel shows all text, all form fields, all sidebar items with correct contrast. 2) Switching to Elite Mode (once unlocked) shows deeper blacks, richer purples, luminous accents, and the Elite glow on key UI. 3) The 5-mode toggle renders correctly on both mobile and desktop. 4) pa11y-ci passes WCAG 2.2 AA on all 17 admin routes in all 5 theme variants. 5) No FOUC — the theme-init script applies the correct palette class before React hydrates.

PART IV · CHAPTER 6

3D Background Enhancements — From CSS Mascots to Immersive WebGL

The current 3D system is a hybrid of three rendering tiers: pure-CSS animated shapes (hero-3d.tsx, nx-3d-scene.tsx, nx-3d-character.tsx — collectively ~80 KB of source that renders floating geometric shapes via CSS transforms), a Canvas2D background (background-scene.tsx — a tech-node graph with 30 nodes, 8 data streams, 2 ripple pulses, capped at 30fps), and one real WebGL scene (nx-three-scene.tsx — Three.js loaded at runtime from a CDN, rendering a wireframe TorusKnot trio, 3 wireframe spheres, a floating icosahedron, and a 1200-particle field with mouse parallax). This chapter specifies the upgrade to a proper @react-three/fiber + drei + postprocessing stack with GLTF character assets, HDRI lighting, and adaptive quality tiering.

Library selection rationale

The recommended stack installs three npm dependencies: @react-three/fiber (the React renderer for Three.js), @react-three/drei (helper components — OrbitControls, Environment, useGLTF, ContactShadows, Float, Sparkles), and @react-three/postprocessing (Bloom, ChromaticAberration, Vignette, Noise). Total gzipped addition to the client bundle: ~280 KB. R3F is preferred over vanilla Three.js because its declarative scene graph reconciles with React, enabling the same component composition patterns the rest of the site already uses. Drei removes 80% of the boilerplate (environment maps, GLTF loading, shadow setup). Postprocessing is what separates "3D" from "high-end 3D" — Bloom alone transforms the visual quality of every emissive material in the scene.

SceneProvider architecture

A new `<SceneProvider>` component wraps the entire site. It hosts a single `<Canvas>` and exposes a context for child components to mount scenes. When the route changes (Next.js App Router), framer-motion's AnimatePresence cross-fades between the outgoing scene and the incoming scene over 1.2 seconds. The Canvas itself never unmounts — only its children swap. This keeps WebGL context alive across navigation, eliminating the cold-start cost (~400ms on mid-tier mobile) of re-initializing Three.js on every page. The Canvas uses `gl={{ antialias: true, powerPreference: "high-performance" }}`, `dpr={[1, 1.75]}` (capped to avoid melting laptop GPUs), and `frameloop="demand"` when the scene is idle.

Performance budget

Hard ceilings enforced in CI via a bundle-size check and a Lighthouse CI run on every PR:

Metric	Budget	Enforcement
3D bundle (gzipped, R3F + drei + postprocessing)	≤ 280 KB	bundlesize check in CI

Metric	Budget	Enforcement
3D scene JS (per-page, gzipped)	≤ 80 KB	per-route bundle analyzer
GLTF asset (per character, compressed)	≤ 150 KB	asset-size pre-commit hook
TBT impact on hero pages	≤ 50 ms	Lighthouse CI
LCP on hero pages (with 3D)	≤ 2.0 s	Lighthouse CI
Frame rate on M1 baseline	≥ 60 fps	Playwright perf test
Frame rate on iPhone 12 baseline	≥ 30 fps	BrowserStack perf test
DPR cap	1.75	Canvas prop

Postprocessing stack

The postprocessing chain applies five effects in order: Bloom (intensity 0.8, luminanceThreshold 0.6, luminanceSmoothing 0.4), ChromaticAberration (offset 0.002, radialModulation true), Vignette (darkness 0.4, offset 0.3), Noise (opacity 0.04), and a custom HologramScan effect for Elite Mode that adds a faint horizontal scan-line. All effects are skippable — the SceneProvider checks `prefers-reduced-motion` and either disables postprocessing entirely or falls back to a single static frame. The Bloom intensity adapts to the page's mood (high on hero pages, low on content pages) via a context prop.

Particle system upgrade

The current 1200-particle CPU field is replaced by a GPU-instanced 5000-particle system using a custom ShaderMaterial. Each particle has a per-instance seed (vec3 attribute) that drives curl-noise displacement in the vertex shader — producing organic, non-repeating motion. The fragment shader renders each particle as a soft radial gradient with additive blending. On low-tier devices (detected via the adaptive quality hook below), the particle count drops to 1500 and the curl-noise octave count drops from 3 to 2.

Adaptive quality tiering

A `useAdaptiveQuality()` hook runs once on mount and returns a quality tier from 0 to 3. Tier detection combines three signals: `navigator.deviceMemory` (where available), `navigator.hardwareConcurrency`, and a real-time frame-time measurement over the first 60 frames after mount. The hook returns immediately with a preliminary tier, then re-fines after measurement. The SceneProvider passes the tier down via context, and each scene component scales its particle count, postprocessing intensity, and shadow map resolution accordingly. Tier 0 = CSS mascots only (no WebGL), tier 1 = WebGL with no postprocessing, tier 2 = WebGL with postprocessing and 1500 particles, tier 3 = full quality.

TSX

```tsx
// src/components/3d/use-adaptive-quality.ts
import { useEffect, useState } from "react";

export function useAdaptiveQuality() {
  const [tier, setTier] = useState<number>(2); // optimistic default
  useEffect(() => {
    const mem = (navigator as any).deviceMemory ?? 4;
    const cores = navigator.hardwareConcurrency ?? 4;
    const reduceMotion = matchMedia("(prefers-reduced-motion: reduce)").matches;
    if (reduceMotion) return setTier(0);
    // Preliminary tier
    let preliminary = 3;
    if (mem <= 2 || cores <= 2) preliminary = 1;
    else if (mem <= 4 || cores <= 4) preliminary = 2;
    setTier(preliminary);
    // Refine via frame timing
    let frames = 0, slowFrames = 0;
    const start = performance.now();
    const interval = setInterval(() => {
      frames++;
      const now = performance.now();
      const dt = now - start;
      if (dt > 1000 / 30 && frames > 5) slowFrames++;
      if (frames >= 60) {
        clearInterval(interval);
        if (slowFrames / 60 > 0.3) setTier(t => Math.max(0, t - 1));
      }
    }, 16);
    return () => clearInterval(interval);
  }, []);
  return tier;
}
```

Fallback ladder

Every 3D scene component implements a three-tier fallback: WebGL (tier 1+), CSS-only mascots (tier 0 or WebGL unavailable), static gradient (CSS mascots failed to mount within 2 seconds). The fallback is automatic — the SceneProvider's Suspense boundary catches the WebGL initialization error and renders the CSS mascot in its place. If the CSS mascot itself throws (extremely rare), the boundary's fallback is a simple div with the brand gradient. This guarantees no visitor ever sees a blank hero section.

PART V · CHAPTER 7

3D Character System — Per-Page Avatars Tied to Content

The 3D character system is the heart of the storytelling platform. Every page features one hero 3D character (or immersive environment) whose narrative role matches the page's content. Characters are not decoration — they are guides that welcome the visitor, react to scroll and cursor, speak on click, and remember the visitor across sessions. This chapter specifies the full character roster, the rig spec, the animation set, the interaction model, and the production pipeline.

Character roster — page mapping

Ten characters, each tied to a page (or page group), each with a distinct silhouette and personality. All characters share a common humanoid base (Ready Player Me low-poly mesh, ~80 KB after Draco compression) with custom geometry accessories that signal their role.

Page	Character	Visual	Role
Home	The Architect	Humanoid assembling floating UI panels	Greets visitor, builds the skyline
Services	The Craftsman	Per-service variant (SEO=spider, DevOps=pipeline, Brand=chromebot)	Shows the tools of each craft
Solutions	The Strategist	Chess-piece figure on a holographic board	Shows the plays that win
About	The Founder	Single character with 4 office skyline backdrops	Tells the origin story
Case Studies	The Explorer	Figure traversing a data landscape	Shows the trophies earned
Blog	The Scribe	Figure writing in 3D space with glowing ink	Shares knowledge
Pricing	The Curator	Figure arranging tier cards on pedestals	Curates the value ladder
Contact	The Messenger	Figure dispatching signals skyward	Sends the visitor's message
Careers	The Pioneer	Figure constructing a building	Invites to build together
Cities	The Navigator	Figure with a slowly rotating globe	Orients across regions

Rig spec

All ten characters share the Ready Player Me humanoid base mesh (low-poly variant, 14k triangles). Custom accessories — hat, tool, weapon, gesture prop — are modeled as separate GLTF meshes (~5k triangles each) attached to the hand or head bone. The character GLTF is Draco-compressed (~80-150 KB total per character) and shipped from the Cloudflare R2 CDN with a 1-year cache. The skeleton is the standard RPM Humanoid rig (65 bones), enabling reuse of any Mixamo or manual animation. Materials use the MeshStandardMaterial with metalness 0.3, roughness 0.6 — catching the HDRI environment without looking plastic. Each character has a single 256×256 emissive texture for the "face" (eyes, mouth, brand-tinted aura) that glows under the Bloom postprocessing.

Animation set

Four animation states per character, all driven by the RPM skeleton and blended with cross-fade transitions (200ms) in R3F's useAnimations hook:

- **Idle** — subtle breathing motion, occasional head turn, weight shift every 4-6 seconds. Loop time 12 seconds. This is the default state when the character is in view but not being interacted with.
- **Hover** — character turns to face the cursor, eyes track cursor position, slight body lean toward cursor. Triggered when the pointer enters the character's bounding box.
- **Scroll-react** — character's pose changes as the user scrolls: at top of page the character is "looking up" (anticipatory), in middle "engaged" (working pose), at bottom "satisfied" (gesture of completion). The pose blends smoothly across three scroll keypoints.
- **Click-react** — character performs a unique "talk" animation: waves, points, nods, or bows depending on personality. Accompanied by a TTS voice line (see Interaction Model below). Cooldown 3 seconds to prevent spam.

Interaction model

Each character responds to three input types: cursor position (hover and parallax), scroll position (pose blending), and click (voice line + animation). Cursor parallax uses a 0.05 lerp factor — the character's head and shoulders rotate up to ±8 degrees toward the cursor with a 200ms lag, producing a subtle "the character is watching me" effect. Scroll-driven pose changes are computed from the character's bounding rect relative to the viewport — three keypoints at top, middle, bottom of the viewport. Click triggers a randomly-selected voice line from a per-character pool of 8-12 lines, synthesized via the z-ai-web-dev-sdk TTS endpoint and cached in IndexedDB on first play. The voice is branded — same voice across all characters, with per-character pitch and rate offsets so each character sounds distinct but recognizable.

Production pipeline

Phase 1 ships three characters (Architect, Craftsman, Strategist) using the RPM base with manually-modeled accessories. Phase 2 adds the remaining seven. The pipeline: (1) generate base mesh on readyplayer.me (free, browser-based), (2) model accessories in Blender (one day per accessory), (3) bake materials and export as Draco-compressed GLTF, (4) author 4 animation clips in Mixamo (free for <2k keyframes per clip), (5) import into the React component via useGLTF and useAnimations, (6) wire interactions in the shared `<Character/>`

wrapper. Total per-character budget: 2 designer-days + 1 developer-day. Total for all 10 characters: ~30 person-days spread across Phases 1-2.

Accessibility & reduced motion

Every character respects `prefers-reduced-motion`. When set, the character renders in a static "portrait" pose (no idle, no parallax, no scroll-react). Click still triggers a voice line (TTS is not motion). Every character is also `aria-hidden="true"` by default — they are decorative guides, not content. The voice-line button is a separate `<button>` with an `aria-label` like "Hear what The Architect says" that appears when the character is focused or hovered.

PART VI · CHAPTER 8

Text Box Redesign — Futuristic 3D-Style UI Component System

Every text-input and text-display component on the site gets a unified redesign: frosted-glass surfaces, 3D extrusion via CSS transforms, magenta focus glow, character-driven microinteractions, and WCAG 2.2 AA accessibility. The system extends shadcn/ui's existing Input and Textarea primitives with a new NxInput wrapper family — `NxInput`, `NxTextarea`, `NxSearchInput`, `NxEmailInput`, `NxPasswordInput`, `NxNumberInput`, `NxOTPInput`, `NxCodeBlock`, `NxCallout`, `NxStatCard`, `NxBlockquote`.

Visual language

The design language is "frosted glass over deep space". Every input surface uses `background: rgba(255, 255, 255, 0.04)` (4% white over the dark base) with a 1px hairline border `rgba(255, 83, 169, 0.2)` (20% magenta). On hover, the border opacity rises to 0.5 and the surface lifts 2px via `transform: translateY(-2px)`. On focus, the border solidifies to magenta, an outer glow (`box-shadow: 0 0 0 3px rgba(255, 83, 169, 0.2)`) appears, and the character animation triggers a " the character is looking at this field" head turn. Depth is achieved via a parent `perspective(1000px)` and a child `transform: translateZ(8px)` on the input — producing a subtle 3D extrusion that reads as "this field is part of the 3D world, not pasted on top of it".

State matrix

State	Surface	Border	Transform	Extra
Default	rgba(255,255,255,0.04)	rgba(255,83,169,0.2) 1px	translateZ(8px)	—
Hover	rgba(255,255,255,0.06)	rgba(255,83,169,0.5) 1px	translateY(-2px) translateZ(10px)	Cursor: pointer
Focus	rgba(255,83,169,0.08)	#FF53A9 1.5px	translateY(-2px) translateZ(12px)	box-shadow: 0 0 0 3px rgba(255,83,169,0.2)
Error	rgba(248,113,113,0.08)	#F87171 1.5px	translateX(0)	Shake animation 200ms
Disabled	rgba(255,255,255,0.02)	rgba(255,255,255,0.05) 1px	translateZ(4px)	opacity 0.5, grayscale
Filled	rgba(255,255,255,0.05)	rgba(255,83,169,0.3) 1px	translateZ(8px)	Check icon right

Microinteractions

Three microinteractions distinguish the NxInput family from a generic styled input. First, the label floats: when the field is empty, the label sits inside the field as placeholder text; on focus or when filled, the label slides up 12px and shrinks from 14px to 11px over 200ms with an ease-out curve. Second, the character counter (where applicable — Textarea, EmailInput with character limit) slides in from the right on focus, displaying "0 / 500" in mono font with a magenta progress arc that fills as the user types. Third, the validation checkmark: when a field passes async validation (e.g. email format check, username availability), a 16px magenta check icon scales in from 0 with a 200ms spring animation.

CodeBlock — special-case component

The NxCodeBlock component renders syntax-highlighted code with a copy button and a language badge. Syntax highlighting uses Shiki (server-side, zero client JS) — the code is highlighted at build time and shipped as semantic HTML with inline styles. The copy button is a 24×24 magenta icon top-right that, on click, copies the raw code to clipboard and morphs into a check icon for 1.5 seconds. The language badge is a small mono-font pill in the top-left, magenta on dark, displaying the language name uppercased. The block itself uses the same frosted-glass surface as other NxInput components but with monospace font and 14px line-height to optimize readability of code.

Accessibility

WCAG 2.2 AA is the baseline. Every NxInput achieves 4.5:1 contrast between text and background. The focus ring is 2px solid magenta with a 3px outer glow — visible against any background, never relying on color alone. Touch targets are 44×44px minimum. Every input wires `aria-invalid`, `aria-describedby` (for error text), and `aria-label` (when no visible label). The shake animation on error respects `prefers-reduced-motion` — instead of shaking, the field shows a 200ms red border pulse. Keyboard navigation: Tab moves between fields, Enter submits forms, Esc clears the current field. Screen readers announce label, current value, and validation status in that order.

Implementation — extending shadcn/ui

The NxInput family extends shadcn/ui's Input, Textarea, and Form components. The wrapper adds three things: the 3D extrusion wrapper (a parent div with perspective and a child with translateZ), the focus-glow box-shadow, and the floating-label animation. The underlying shadcn/ui Input keeps its data-binding and form-integration behavior intact. A single `cn()` utility merges the new className with the shadcn default. The result: any existing form on the site can opt into the new design by swapping `Input` for `NxInput` — no other code changes required.

TSX

```tsx
// src/components/ui/nx-input.tsx
"use client";
import * as React from "react";
import { cn } from "@/lib/utils";

export interface NxInputProps extends React.InputHTMLAttributes<HTMLInputElement> {
  label?: string;
  error?: string;
}

export const NxInput = React.forwardRef<HTMLInputElement, NxInputProps>(
  ({ className, label, error, id, ...props }, ref) => {
    const inputId = id || React.useId();
    return (
      <div className="nx-input-wrapper" style={{ perspective: "1000px" }}>
        {label && (
          <label htmlFor={inputId} className="nx-input-label">
            {label}
          </label>
        )}
        <div style={{ transformStyle: "preserve-3d", transform: "translateZ(8px)" }}>
          <input
            id={inputId}
            ref={ref}
            aria-invalid={!!error}
            className={cn(
              "nx-input",
              error && "nx-input-error",
              className
            )}
            {...props}
          />
        </div>
        {error && <p className="nx-input-error-text">{error}</p>}
      </div>
    );
  }
);
NxInput.displayName = "NxInput";
```

Token table — paste into Figma

Surface rgba(255,255,255,0.04) · Border 1px rgba(255,83,169,0.2) · Border-focus 1.5px #FF53A9 · Glow 0 0 0 3px rgba(255,83,169,0.2) · Label-float 14px→11px, 0→-12px · Counter Mono 11px rgba(255,83,169,1) · Check 16px #FF53A9 spring 200ms · Padding 12px 16px · Radius 8px · Font NotoSans 14px/20px · Touch-target 44×44px min · Focus-ring 2px #FF53A9 + 3px glow.

PART VII · CHAPTER 9

Immersive Storytelling — Narrative Architecture Across Pages

The storytelling layer ties every element of the redesign together. Without it, the 3D characters are decoration and the redesigned text boxes are isolated components. With it, the site becomes a cohesive journey: the visitor is a founder exploring a digital city — ClickTake City — and every page is a district with its own guide. This chapter defines the meta-narrative, the page-by-page arc, the transition choreography, and the persistence layer that makes the journey feel continuous across sessions.

Meta-narrative — The Visitor's Journey

The visitor arrives at ClickTake City as a founder with an idea. They are greeted at the city gates (Home) by The Architect, who shows them the skyline of what could be built. They visit the workshop district (Services) where The Craftsman shows them the tools. They enter the war room (Solutions) where The Strategist shows them the plays. They browse the trophy gallery (Case Studies) where The Explorer recounts past victories. They enter the boutique (Pricing) where The Curator arranges the value ladder. They visit city hall (About) where The Founder tells the origin story. They stop by the library (Blog) where The Scribe shares knowledge. They climb the comm tower (Contact) where The Messenger dispatches their signal skyward. They pass the construction site (Careers) where The Pioneer invites them to build. They meet The Navigator (Cities) at the city's crossroads, who orients them across the regions ClickTake serves.

Page-by-page narrative arc

Page	District	Character	Narrative Beat
Home	City Gates	The Architect	Arrival — "Welcome, founder. Let me show you the skyline."
Services	Workshop	The Craftsman	Tools — "These are the instruments we build with."
Solutions	War Room	The Strategist	Plays — "Every founder needs a playbook."
Case Studies	Trophy Gallery	The Explorer	Trophies — "These are the journeys we've taken."
Pricing	Boutique	The Curator	Value — "Choose the tier that fits your ambition."
About	City Hall	The Founder	Origin — "Here is how ClickTake City came to be."
Blog	Library	The Scribe	Knowledge — "These are the lessons we've written down."

Page	District	Character	Narrative Beat
Contact	Comm Tower	The Messenger	Signal — "Tell us where to send the signal."
Careers	Construction Site	The Pioneer	Build — "We're building something. Join us."
Cities	Crossroads	The Navigator	Orient — "Four cities, one studio."

Transition choreography

When the visitor navigates between pages, the outgoing character waves goodbye and shrinks to 0.3× scale over 0.6 seconds, then fades out. Simultaneously, the incoming character materializes via a particle-assembly effect: 500 particles spawn at random positions within the character's bounding box and converge to the character mesh over 1.0 seconds, with the mesh opacity rising from 0 to 1 over the final 0.4 seconds. The total transition is 1.2 seconds — long enough to feel intentional, short enough to not frustrate frequent navigators. The transition is skipped on back-button navigation (the visitor is "returning", not "arriving") and on reduced-motion (instant swap with a simple fade).

Scroll-driven narrative

Within each page, the character's pose changes as the visitor scrolls through sections. Three keypoints — top, middle, bottom of the page — drive the pose blend. At the top (anticipatory), the character faces the visitor with a "let me show you" expression. In the middle (engaged), the character turns to face the content, gesturing toward the section in view. At the bottom (satisfied), the character turns back to the visitor with a "what next?" expression. The pose blend uses the same useScroll hook from @react-three/drei that drives the scroll-react animation in Chapter 7 — the two systems are unified, not duplicated.

Voice-over — optional TTS narration

Each page can optionally play a 3-5 second TTS-narrated intro when the character materializes. The narration is a single sentence in the character's voice, synthesized via the z-ai-web-dev-sdk TTS endpoint on first visit and cached in IndexedDB. Examples: The Architect on Home — " Welcome, founder. Let me show you the skyline." The Craftsman on Services — "These are the instruments we build with." The Strategist on Solutions — "Every founder needs a playbook." Voice-over is opt-in — the visitor enables it via a toggle in the navbar (off by default). Once enabled, it plays once per page per session (not on every navigation back to the same page). A small "skip" button appears bottom-right while narration plays.

Persistence — the character remembers

A small localStorage entry (`clicktake-journey`) tracks the visitor's journey: which districts they've visited, how many sessions, last visit date. On session 2+, the home character's voice line changes from "Welcome, founder" to "Welcome back. Let's pick up where we left off." If the visitor has visited 5+ districts, a small "Founder's Path" badge appears in the navbar showing progress through the 10 districts. If the visitor has visited all 10, an "Elite Mode unlocked" modal appears — turning the storytelling journey into the Elite Mode opt-in mechanic. This is how Elite Mode adoption reaches the 5% Q1 KPI: the journey rewards completion with Elite access.

Narrative flow diagram (text representation)

FLOW

```
                    ┌─────────────────┐
                    │ ARRIVAL (Home)  │
                    │ The Architect   │
                    └────────┬────────┘
                             │
           ┌─────────────────┼──────────────────┐
           ▼                 ▼                  ▼
     ┌──────────┐      ┌──────────┐      ┌──────────┐
     │ Services │      │Solutions │      │ Pricing  │
     │Craftsman │      │Strategist│      │ Curator  │
     └────┬─────┘      └────┬─────┘      └────┬─────┘
          │                 │                 │
          └────────┬────────┴────────┬────────┘
                   ▼                 ▼
            ┌──────────┐      ┌──────────┐
            │Case Study│      │  About   │
            │ Explorer │      │ Founder  │
            └────┬─────┘      └────┬─────┘
                 │                 │
                 ▼                 ▼
            ┌──────────┐      ┌──────────┐
            │   Blog   │      │  Cities  │
            │  Scribe  │      │Navigator │
            └────┬─────┘      └────┬─────┘
                 │                 │
                 └────────┬────────┘
                          ▼
                    ┌──────────┐
                    │ Contact  │
                    │Messenger │
                    └────┬─────┘
                         ▼
                    ┌──────────┐
                    │ Careers  │
                    │ Pioneer  │ ──► Elite Mode unlocked
                    └──────────┘
```

PART VIII · CHAPTER 10

Implementation Roadmap

A six-phase, 18-week implementation plan. Each phase has a clear deliverable, an owner suggestion, a definition-of-done, and a risk profile. Phase gates allow pause points if team bandwidth tightens — a smaller team can ship Phases 0-3 (SEO wins plus the hero 3D experience on five pages) in 9 weeks and still deliver a visible transformation. The full 18-week plan delivers the complete 3D storytelling platform with Elite Mode, voice-over, and persistence.

Phase overview (Gantt)

Phase	Weeks	Owner	Deliverable	Risk
Phase 0 — SEO/GEO quick wins	1	FE + DevOps	llms.txt, RSS, FAQ schema, robots.txt cleanup, Product schema, Vercel Analytics	Low
Phase 1 — Foundation	2-3	FE + 3D	R3F/drei/postprocessing install, SceneProvider, adaptive quality hook, Elite Mode tokens, 5-mode toggle	Medium — bundle size
Phase 2 — Character system	4-6	3D + Design	3 base characters (Architect, Craftsman, Strategist), idle/hover/scroll/click animations, RPM pipeline	Medium — asset production
Phase 3 — Page rollout	7-9	FE + 3D	Characters deployed to 10 pages (1 per week, A/B tested), voice-over on 3 hero pages	Medium — perf regression per page
Phase 4 — UI system v2	10-11	Design + FE	NxInput family (10 components), CodeBlock, Callout, StatCard migration, RSS extension to /research/	Low
Phase 5 — Storytelling layer	12-13	FE + 3D	Transitions, scroll-narrative, TTS voice-over, persistence, Elite unlock modal	Medium — TTS cost
Phase 6 — Polish & perf	14-16	FE + DevOps	Bundle audit, CWV optimization, pa11y-ci in CI, Elite Mode premium gate	Medium — scope creep
Phase 7 — Launch	17-18	All	Staged rollout (10% → 50% → 100%), GEO monitoring dashboard, post-launch retro	Low

Phase 0 — SEO/GEO quick wins (week 1)

Ship all 7 quick-win SEO/GEO gaps in one week. Full code patches in Appendix A. Deliverable: a single PR that adds llms.txt, RSS feed, FAQ schema on home, deletes public/robots.txt, adds Product schema on pricing, Review schema on testimonials, and migrates admin @import fonts to next/font. Verification: Google Rich Results Test passes on all 5 modified pages, Schema.org validator 100% pass, Vercel Analytics dashboard live. Owner: 1 frontend engineer + 1 DevOps engineer. Risk: low — all changes are additive or deletions, no breaking changes to existing routes.

Phase 1 — Foundation (weeks 2-3)

Install @react-three/fiber, @react-three/drei, @react-three/postprocessing. Build `<SceneProvider>` with single Canvas, AnimatePresence scene swapping, adaptive quality hook, postprocessing chain. Build the Elite Mode token cascade from Chapter 5 and upgrade theme-toggle to 5-mode segmented control. Definition-of-done: Lighthouse CI passes on a five-page subset, 3D bundle ≤ 280 KB gzipped, Elite Mode toggle works on admin and public site. Owner: 1 frontend engineer (lead) + 1 3D engineer. Risk: medium — bundle size discipline required; the postprocessing chain alone is ~80 KB and must be lazy-loaded only when WebGL is available.

Phase 2 — Character system (weeks 4-6)

Rig 3 base characters (Architect, Craftsman, Strategist) using Ready Player Me base + Blender-modeled accessories. Author 4 animation clips per character (idle, hover, scroll-react, click-react) in Mixamo. Build the shared `<Character/>` wrapper with parallax, scroll, and click handlers. Definition-of-done: all 3 characters render at 60fps on M1 baseline, 30fps on iPhone 12, ≤ 150 KB per character GLTF. Owner: 1 3D engineer + 1 designer (modeling). Risk: medium — character asset production is the most uncertain timeline component; if RPM base doesn't meet quality bar, fall back to custom-modeled low-poly meshes (adds 1 week).

Phase 3 — Page rollout (weeks 7-9)

Deploy characters to all 10 pages — one per week, A/B tested (50% see new 3D character, 50% see current CSS mascot). Measure bounce rate, avg session duration, conversion on /contact. Roll forward if 3D variant doesn't regress any metric by more than 5%. Add voice-over on 3 hero pages (Home, Services, About). Definition-of-done: 10 pages live with 3D characters, A/B test results documented, voice-over on 3 pages. Owner: 1 frontend engineer + 1 3D engineer. Risk: medium — perf regression per page is the biggest risk; mitigate by running Lighthouse CI on every page-add PR.

Phase 4 — UI system v2 (weeks 10-11)

Build NxInput family (10 components), migrate all forms (contact, newsletter, login, admin CMS) to new components. Add CodeBlock, Callout, StatCard components. Build the /research/ section with the three original-research assets (State of AI-Search 2026, GEO Benchmark, Client Metrics Dashboard). Definition-of-done: every form uses NxInput family, pa11y-ci passes on all forms in all 5 theme variants, /research/ section live with 3 downloadable assets. Owner: 1 designer + 1 frontend engineer. Risk: low — component-swap work, no architectural changes.

Phase 5 — Storytelling layer (weeks 12-13)

Wire transitions between pages (particle-assembly materialization), scroll-driven pose blending, TTS voice-over for all 10 characters (8-12 lines each), persistence via localStorage, Elite unlock modal. Definition-of-done: full storytelling journey works end-to-end, voice-over toggle in navbar, Elite unlock modal appears after 10 districts visited. Owner: 1 frontend engineer + 1 3D engineer. Risk: medium — TTS cost via z-ai-web-dev-sdk scales with usage; mitigate by caching every line in IndexedDB on first play and only re-synthesizing if the line text changes.

Phase 6 — Polish & performance (weeks 14-16)

Bundle audit — verify 3D bundle ≤ 280 KB, per-page ≤ 80 KB. CWV optimization — chase LCP < 2.0s on every hero page, TBT < 150ms, CLS < 0.05. pa11y-ci in CI on all 17 admin routes × 5 theme variants. Elite Mode premium gate — content modules gated behind Elite unlock. Definition-of-done: Lighthouse CI green on all routes, pa11y-ci green on all variants, Elite Mode premium content live. Owner: 1 frontend engineer + 1 DevOps engineer. Risk: medium — scope creep risk on Elite Mode premium content; mitigate by defining MVP as visual layer only and deferring premium content to v2.

Phase 7 — Launch (weeks 17-18)

Staged rollout: 10% of traffic in week 17 (Vercel Edge Config flag), 50% mid-week, 100% by end of week 18. GEO monitoring dashboard live (Perplexity API + Google Search Console AI queries + manual weekly ChatGPT/Perplexity prompts). Post-launch retro at end of week 18. Definition-of-done: 100% rollout, GEO dashboard showing baseline metrics, retro document published. Owner: full team. Risk: low — staged rollout catches any production-only issues before full exposure.

PART IX · CHAPTER 11

KPIs, Measurement & Success Criteria

Success is measured across four dimensions: SEO (organic visibility), GEO (AI-search visibility), 3D performance (Core Web Vitals), and UX (engagement and conversion). Each dimension has hard targets, measurement infrastructure, and a reporting cadence. The measurement infrastructure installs in Phase 0 (Vercel Analytics + Speed Insights) and Phase 4 (custom GEO dashboard + brand-mention alerts).

SEO KPIs

Organic traffic is the north-star SEO metric. The 6-month target is +40% organic sessions measured in Google Search Console (GSC) — comparing the 6 months post-launch to the 6 months pre-launch. Secondary KPIs: non-brand keyword coverage +60% (measured as count of distinct queries in GSC that don't contain "clicktake"); crawl budget waste < 15% (measured as count of low-value URLs crawled — anything in /api/ or /admin/ that should be disallowed); schema validation 100% pass in Schema.org validator. Reporting cadence: weekly snapshot, monthly deep-dive.

GEO KPIs

Brand mentions in AI answer engines is the north-star GEO metric. The 6-month target is 0 → 50+ mentions per month across ChatGPT Search, Perplexity, Google AI Overviews, and Claude. Measurement: weekly Perplexity API query for "ClickTake" plus 20 category queries; weekly ChatGPT Search prompts for the same 20 queries; weekly Claude prompts (manual, no API yet); weekly Google Search Console AI Overview filter. Secondary KPIs: llms.txt fetches tracked in Vercel Analytics; ai.* referral traffic +200% (measured in Vercel Analytics). Reporting cadence: weekly snapshot in the custom GEO dashboard.

3D performance KPIs

Core Web Vitals on hero pages is the north-star performance metric. Targets: LCP < 2.0s, TBT < 150ms, CLS < 0.05. Measurement: Vercel Speed Insights (RUM from real users) + Lighthouse CI (synthetic, runs on every PR). Secondary KPIs: 3D bundle ≤ 280 KB gzipped (enforced via bundlesize in CI); frame rate ≥ 60fps on M1 baseline, ≥ 30fps on iPhone 12 baseline (measured via Playwright + BrowserStack perf tests, run nightly). Reporting cadence: per-PR (Lighthouse CI blocking), daily (RUM dashboard), weekly (perf regression review).

UX KPIs

Bounce rate reduction on pages with 3D characters is the north-star UX metric. Target: -15% bounce rate on the 10 character pages (measured in Vercel Analytics). Secondary KPIs: avg session duration +30%; conversion rate on /contact +20%; Elite Mode adoption 5% of returning users in Q1 (measured as count of users with localStorage `clicktake-journey` containing all 10 districts, divided by total returning users). Reporting cadence: weekly snapshot, monthly deep-dive with cohort analysis.

KPI summary table

Dimension	KPI	Target	Source	Cadence
SEO	Organic sessions	+40% in 6 mo	Google Search Console	Weekly
SEO	Non-brand keyword coverage	+60%	GSC queries report	Monthly
SEO	Schema validation pass	100%	Schema.org validator	Per-PR
GEO	AI-search brand mentions	0 → 50+/mo	Perplexity API + manual	Weekly
GEO	llms.txt fetches	Tracked, trending up	Vercel Analytics	Weekly
GEO	ai.* referral traffic	+200%	Vercel Analytics	Weekly
3D	LCP on hero pages	< 2.0s	Vercel Speed Insights	Daily
3D	TBT	< 150ms	Lighthouse CI	Per-PR
3D	CLS	< 0.05	Lighthouse CI	Per-PR
3D	3D bundle (gzipped)	≤ 280 KB	bundlesize CI	Per-PR
3D	Frame rate on M1	≥ 60 fps	Playwright perf test	Nightly
3D	Frame rate on iPhone 12	≥ 30 fps	BrowserStack perf test	Nightly
UX	Bounce rate on character pages	-15%	Vercel Analytics	Weekly
UX	Avg session duration	+30%	Vercel Analytics	Weekly
UX	Conversion rate on /contact	+20%	Vercel Analytics	Weekly
UX	Elite Mode adoption Q1	5% of returning users	localStorage audit	Monthly

Measurement infrastructure

Three layers. First, Vercel Analytics + Vercel Speed Insights install on day 1 of Phase 0 — free, privacy-first, no consent banner required. Surfaces pageview counts, ai.* referral sources, and Core Web Vitals RUM. Second, PostHog installs in Phase 1 for product analytics (session replay, funnel analysis, feature flags for the staged rollout). Third, a custom GEO dashboard builds in Phase 4 — a Next.js route at /admin/geo (admin-only) that aggregates weekly Perplexity API results, GSC AI Overview data, and manual ChatGPT/Claude prompt results into a single timeline view. The dashboard also runs the brand-mention alert pipeline (Phase 4) — a nightly cron that pings Slack whenever ClickTake is newly cited in a major AI answer engine.

PART X · CHAPTER 12

Risk Register & Open Questions

Eight risks are tracked across the 18-week plan. Each has a mitigation strategy and an owner. Risks are reviewed weekly during the standup and escalated to the steering committee if probability or impact rises. Open questions are listed at the end — these are decisions that need a stakeholder answer before Phase 5.

Risk register

#	Risk	Prob	Impact	Mitigation
1	3D bundle bloat tanks Core Web Vitals	Med	High	Strict 280KB budget enforced via bundlesize CI; adaptive quality tiering degrades to CSS mascots on low-tier devices
2	Accessibility regression from 3D	Low	High	WCAG 2.2 audit per phase; prefers-reduced-motion respect; pa11y-ci in CI on all 17 admin routes × 5 themes
3	Mobile perf on low-end devices	Med	High	GPU tier detection via useAdaptiveQuality hook; graceful degradation to CSS mascots; nightly BrowserStack perf test on iPhone 12
4	GLTF asset production bottleneck	Med	Med	Start with Ready Player Me free base; customize accessories later; pre-built Mixamo animations for the 4 standard states
5	Elite Mode scope creep	High	Med	MVP defined as visual layer only; premium content gating deferred to v2; phase gate at end of Phase 6 to review scope
6	SEO volatility during redesign	Low	High	301-map every URL; keep sitemap.ts stable; deploy SEO fixes in Phase 0 before any 3D work; daily GSC monitoring during Phase 3
7	GEO measurement gap (no analytics today)	High	Med	Install Vercel Analytics Day 1 of Phase 0; build custom GEO dashboard in Phase 4; baseline measurement for 2 weeks before Phase 3
8	Team bandwidth for 18-week timeline	Med	High	Phase gates allow pause points; Phases 0-3 (9 weeks) deliver visible transformation; Phase 4-7 can stretch if needed

Open questions for stakeholders

These decisions need answers from ClickTake leadership before Phase 5 (storytelling layer) begins. Each question affects scope, budget, or timeline.

- **Localization**: Do we localize the site for Arabic (UAE office) and Urdu (Pakistan office) markets? This affects hreflang implementation, content translation budget, and the character voice-over production (each line × 3 languages).
- **Elite Mode monetization**: Is Elite Mode a free loyalty reward (the default in this brief) or a paid tier? If paid, what's the pricing — one-time, monthly, annual? This affects Phase 6 premium gate implementation.
- **Character licensing**: Do we license Ready Player Me (free, attribution required, generic look) or commission original character rigs from a 3D artist (estimated $5-10k per character, exclusive rights, distinctive brand silhouettes)? This affects Phase 2 timeline and budget.
- **Voice-over shipping**: Do we ship TTS voice-over in Phase 5 (per this brief) or defer to v2? TTS via z-ai-web-dev-sdk costs ~$0.001 per line — for 100 lines × 10 characters × ~1000 unique visitors per day = ~$1000/month. Acceptable? Or wait for v2 with human voice talent?
- **Original research cadence**: The /research/ section proposes three assets updated quarterly. Who owns the research — internal team, external contractor, or a hired research lead? This affects Phase 4 and ongoing operational cost.
- **Elite Mode metrics gating**: When a visitor unlocks Elite Mode by visiting all 10 districts, do we also gate by session count (e.g. ≥3 sessions) or time-on-site (e.g. ≥10 minutes)? Affects the unlock modal logic in Phase 5.

> **Recommendation on open questions**
>
> Default to the simpler option for each: skip localization in v1 (single-locale is fine for now), Elite Mode as free loyalty reward, RPM base with custom accessories (the hybrid in this brief), ship TTS voice-over in Phase 5 (the $1000/month cost is acceptable for the engagement lift), internal team owns research (hire a contractor only if Phase 4 slips), gate Elite Mode by district count only (simplest logic). These defaults can be revisited in the Phase 6 review.

APPENDIX A

Quick-Win Code Patches

Copy-paste-ready patches for the 7 Phase 0 SEO/GEO quick wins. Each patch has a file path, the code, and a verification command. All patches are designed to ship as a single PR in week 1.

A.1 — public/llms.txt

Create `public/llms.txt` with the full content drafted in Chapter 4. Verification: `curl https://clicktaketech.com/llms.txt` returns the markdown.

A.2 — Delete public/robots.txt

SHELL

```
git rm public/robots.txt
# Verify
curl -sI https://clicktaketech.com/robots.txt | head -5
# Should return Content-Type: text/plain with the dynamic robots.ts output
# Body should include "Sitemap: https://clicktaketech.com/sitemap.xml"
```

A.3 — RSS feed at /feed.xml

Full code in Gap 5 of Chapter 2. Create `src/app/feed.xml/route.ts`. Verification: `curl https://clicktaketech.com/feed.xml` returns valid Atom XML. Add `<link rel="alternate" type="application/atom+xml" href="/feed.xml" />` to the document head in layout.tsx.

A.4 — Homepage FAQPage schema

Extract the 6 FAQ items from the NxFaq accordion in src/app/page.tsx into a shared constant exported to a JSON-LD builder. Add a script tag with the FAQPage schema below the NxFaq component. Full code in Gap 2 of Chapter 2. Verification: Google Rich Results Test on https://clicktaketech.com/ shows FAQ rich result eligibility.

A.5 — Product + Offer schema on /pricing

TSX

```tsx
// src/app/pricing/page.tsx — add before closing </main>
<script type="application/ld+json" dangerouslySetInnerHTML={{
  __html: JSON.stringify({
    "@context": "https://schema.org",
    "@type": "ItemList",
    itemListElement: PRICING_PLANS.map((plan, i) => ({
      "@type": "ListItem",
      position: i + 1,
      item: {
        "@type": "Product",
        name: plan.name,
        description: plan.description,
        brand: { "@type": "Brand", name: "ClickTake" },
        offers: {
          "@type": "Offer",
          price: plan.price.replace(/[^0-9]/g, ""),
          priceCurrency: "USD",
          availability: "https://schema.org/InStock"
        }
      }
    }))
  })
}} />
```

A.6 — Review schema for testimonials

TSX

```tsx
// src/components/site/nx-testimonials.tsx — add before closing </section>
const reviewSchema = {
  "@context": "https://schema.org",
  "@type": "Product",
  name: "ClickTake Web Development Services",
  aggregateRating: {
    "@type": "AggregateRating",
    ratingValue: "5.0",
    reviewCount: TESTIMONIALS.length,
    bestRating: "5",
    worstRating: "1"
  },
  review: TESTIMONIALS.map(t => ({
    "@type": "Review",
    author: { "@type": "Person", name: t.name },
    reviewBody: t.quote,
    reviewRating: { "@type": "Rating", ratingValue: "5", bestRating: "5" }
  }))
};
// Render as <script type="application/ld+json" ...>
```

A.7 — next/font migration for admin

TSX

```tsx
// src/app/admin/layout.tsx — replace @import in admin-globals.css with next/font
import { Syne, JetBrains_Mono } from "next/font/google";

const syne = Syne({ subsets: ["latin"], weight: ["400", "500", "600", "700", "800"], display: "swap",
variable: "--font-syne" });
const jetbrains = JetBrains_Mono({ subsets: ["latin"], weight: ["400", "500"], display: "swap", variable: "
--font-jetbrains" });

// In the layout JSX:
<body className={`${syne.variable} ${jetbrains.variable} ct-admin`}>
  {children}
</body>

// Then in admin-globals.css, replace:
// @import url("https://fonts.googleapis.com/css2?family=Syne:wght@400;500;600;700;800&family=JetBrains+Mono:
wght@400;500&display=swap");
// With nothing (next/font handles loading). Update font-family declarations to use var(--font-syne) and
var(--font-jetbrains).
```

A.8 — Elite Mode token cascade (partial — full in Chapter 5)

The full Elite Mode CSS token cascade is in Chapter 5. For Phase 0, ship only the foundation: add `data-theme` attribute switching to the html element (replacing the current class-based .dark / .theme-custom system), define the base :root tokens, and migrate the top 10 most-used admin-globals.css rules to read from tokens. Full migration completes in Phase 1.

APPENDIX B

Glossary & References

A working glossary of the 30+ terms used in this brief, plus reference links for deeper reading.

Glossary

Term	Definition
GEO	Generative Engine Optimization — making content discoverable and citeable by AI answer engines (ChatGPT Search, Perplexity, Google AI Overviews, Claude).
SEO	Search Engine Optimization — traditional optimization for crawler-based search engines (Google, Bing).
llms.txt	A markdown-formatted file at /llms.txt proposed by Jeremy Howard (2024) that gives LLMs a curated overview of site content. The GEO equivalent of sitemap.xml.
R3F	@react-three/fiber — the React renderer for Three.js. Enables declarative scene graphs that reconcile with React.
drei	@react-three/drei — helper components for R3F (OrbitControls, Environment, useGLTF, ContactShadows, Float, Sparkles).
postprocessing	@react-three/postprocessing — Bloom, ChromaticAberration, Vignette, Noise effects for R3F scenes.
Draco	Google's 3D geometry compression library. Compresses GLTF meshes by 90%+ with minimal quality loss.
Meshopt	Alternative 3D compression library; complements Draco for animation and morph targets.
Ready Player Me	Cross-game avatar platform; provides free humanoid base meshes with standard skeleton. Used as the character base in this brief.
HDRI	High Dynamic Range Image — used as environment lighting in 3D scenes to produce realistic material reflections.
WCAG 2.2 AA	Web Content Accessibility Guidelines version 2.2, conformance level AA. Requires 4.5:1 contrast for body text, 3:1 for large text.
Core Web Vitals	Google's user-experience metrics: LCP (Largest Contentful Paint), TBT (Total Blocking Time), CLS (Cumulative Layout Shift).
LCP	Largest Contentful Paint — time until the largest visible element renders. Target < 2.5s; this brief targets < 2.0s.
TBT	Total Blocking Time — sum of all long-task blocking periods between FCP and TTI. Target < 200ms; this brief targets < 150ms.
CLS	Cumulative Layout Shift — total visible layout shift. Target < 0.1; this brief targets < 0.05.

Term	Definition
FOUC	Flash of Unstyled Content — the brief flash of default-styled content before the theme initializes. Prevented via the themeInitScript in layout.tsx.
SSR	Server-Side Rendering — Next.js renders HTML on the server, client hydrates. Used for static and dynamic pages.
SSG	Static Site Generation — Next.js renders HTML at build time. Used for blog posts via generateStaticParams.
ISR	Incremental Static Regeneration — SSG with periodic revalidation. Used for city pages (revalidate = 86400).
OpenAPI 3.1	Industry-standard API description format. ClickTake exposes /openapi.json with x402 payment metadata.
MCP	Model Context Protocol — Anthropic's protocol for tool-calling agents. ClickTake exposes /api/mcp and /.well-known/mcp/server-card.json.
x402	HTTP Payment Protocol — enables per-request payments via HTTP 402 status. ClickTake exposes /api/premium returning 402 with payment requirements.
auth.md	WorkOS-spec markdown file at /auth.md describing how AI agents authenticate with a site.
DNS-AID	DNS-based Agent Identification — RFC 9460 SVCB/HTTPS records that publish agent endpoints at the DNS layer.
RFC 8285 Link headers	HTTP Link header spec used to advertise related resources (api-catalog, service-desc, oauth-authorization-server, etc.).
WebMCP	Experimental browser API exposing site tools to AI agents via the MCP protocol. ClickTake implements a WebMCP browser provider (Chrome-only).
shadcn/ui	Component collection built on Radix UI primitives. ClickTake uses 49 shadcn components in src/components/ui/.
TTS	Text-to-Speech — synthesizing audio from text. This brief uses z-ai-web-dev-sdk TTS for character voice lines.
Vercel Analytics	Privacy-first, cookie-less analytics built into Vercel. Free for hobby tier; no consent banner required.
PostHog	Open-source product analytics with session replay, feature flags, and A/B testing. Recommended for Phase 1 install.

References

- **ClickTake live site**: `https://clicktaketech.com`
- **Next.js 16 docs**: `https://nextjs.org/docs`
- **@react-three/fiber docs**: `https://r3f.docs.pmnd.rs`
- **@react-three/drei docs**: `https://drei.docs.pmnd.rs`
- **llms.txt spec**: `https://llmstxt.org`

- **Schema.org**: https://schema.org
- **Web.dev Core Web Vitals**: https://web.dev/vitals
- **Ready Player Me**: https://readyplayer.me
- **Vercel Analytics**: https://vercel.com/analytics
- **WCAG 2.2**: https://www.w3.org/TR/WCAG22

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